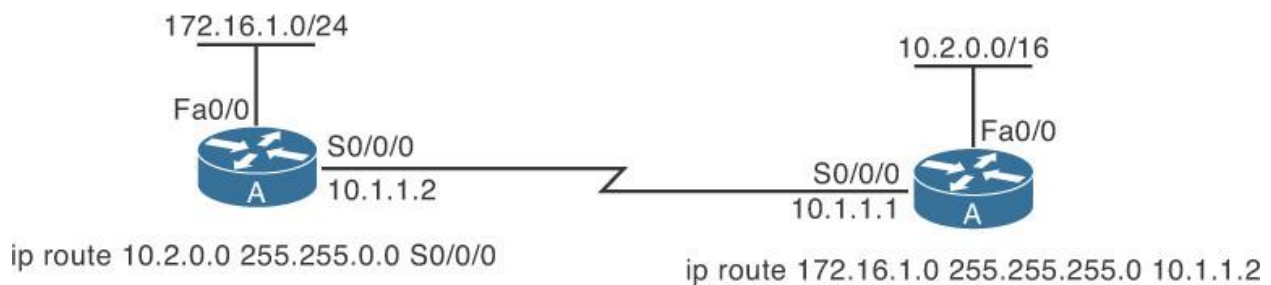


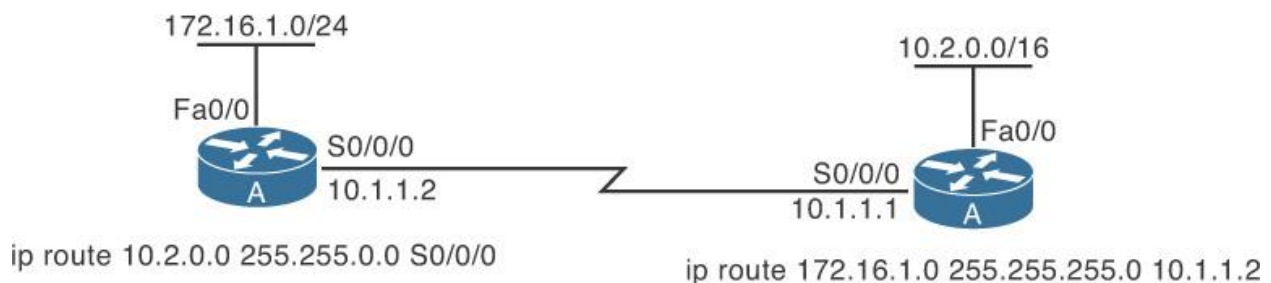
Configure & Verify Static Routing:

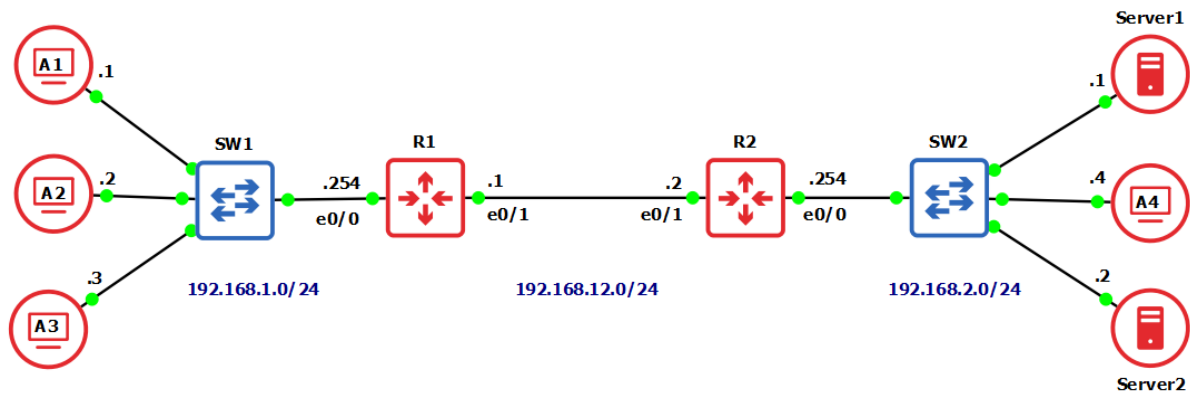
- o When routers learn from a Network administrator, it is called a static routing.
- o In Static Routing administrator manually inputs all the routing table information.
- o Static route tells network devices about exact location & only work in small network.
- o In static routing, we must add all network locations manually or statically in router.
- o If any change occurs in the network, admin is responsible to update it in all routers.
- o Keep in mind, that a static route will have a default administrative distance (AD) of 1.
- o To keep route all time in Routing table, **permanent** keyword to end of route statement.
- o This keyword will keep the route in the table even if the router interface goes down.



Network Route:

- o Network Route used by a router when there is no other known route exists.
- o All the IP with unknown destination address is sent to the default route.
- o Network Route is route for a classful network most entries are network routes.





Configure & Verify Static Routing

```
R1(config)# ip route 0.0.0.0 0.0.0.0 192.168.12.2
```

```
R1(config)# ip route 0.0.0.0 0.0.0.0 ethernet0/1
```

```
R1(config)# ip route 192.168.2.0 255.55.255.0 192.168.12.2
```

```
R1(config)# ip route 192.168.2.0 255.55.255.0 ethernet 0/1
```

```
R1(config)# ip route 192.168.2.1 255.55.255.255 192.168.12.2
```

```
R1(config)# ip route 2.0.0.0 255.0.0.0 192.168.12.2
```

```
R1(config)# ip route 2.0.0.0 255.0.0.0 192.168.21.2 5
```

```
R1(config)# ip route 0.0.0.0 0.0.0.0 192.168.12.2
```

```
R1(config)# ip route 0.0.0.0 0.0.0.0 192.168.21.2 5
```

```
R1# show ip route
```

```
R1# show ip route connected
```

```
R1# show ip route static
```

```
R1# show ip route 2.2.2.2
```

```
R1# show ip route 192.168.2.0
```